NUCLEUS The dense center of an atom where protons and neutrons are found.

OPTICAL CLOCK A special type of clock that uses light and single atoms to keep highly accurate time.

ORBITAL Spherical layer, or shell, around the nucleus of an atom where electrons can be found.

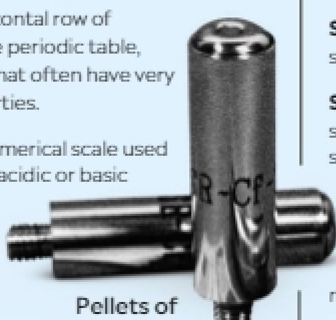
ORE Naturally occurring rock from which useful minerals can be extracted.

ORGANIC CHEMISTRY A branch of chemistry that deals with the chemistry of carbon and its compounds.

OXIDE A compound that forms when an element combines with oxygen.

PERIOD A horizontal row of elements on the periodic table, with elements that often have very different properties.

PH SCALE A numerical scale used to describe the acidic or basic properties of a substance. Less than 7 on the scale is described as acidic, above 7 as basic, and 7 as neutral.



Pellets of radioactive californium

PHOTOSYNTHESIS Naturally occurring chemical reaction that plants use to turn light energy into chemical energy.

Fluorite, a mineral containing fluorine

POROUS A material with tiny holes through which liquids can pass.

PROTON A positively charged particle in the nucleus of an atom.

RADIATION Energy released, usually in the form of waves or particles.

RADIOACTIVITY A process that occurs when an atom is unstable because the protons and neutrons in the nucleus do not stick together.

REACTIVITY A substance's tendency to undergo chemical reactions.

SALT A compound that forms when an acid reacts with an alkali. Sodium chloride is the most familiar example of a salt.

SEMICONDUCTOR A material that conducts electricity better than an insulator but not as well as a metal.

SOLUBLE A substance that dissolves in a solvent (usually water).

STATES OF MATTER The three common states of matter are solid, liquid, or gas. In solids, particles are bound to each other, so they remain in fixed positions. In liquids, particles are loosely attached to each other and move freely. In gases, particles are not attached to each other and can move away.

SUBATOMIC PARTICLE A particle that makes up an atom. These include protons, neutrons, and electrons.

SUPERALLOY A combination of metals that can withstand extreme temperature and pressure.

SUPER-HEAVY ELEMENTS Elements with the atomic number 104 or higher.

SYNTHETIC ELEMENT An element created artificially in a laboratory.

TEMPERATURE How hot or cold something is as measured on a defined scale.

TOXICITY A measure of the poisonous properties of a substance.

TRANSURANIUM ELEMENTS Elements with an atomic number higher than that of uranium (92).

ULTRAVIOLET RAYS Invisible electromagnetic radiation with very short wavelengths. It is called ultraviolet because it is beyond the violet end of visible light on the electromagnetic spectrum.

Jet engine with inner blades made of a rhenium superalloy

VAPOR A mixture of gaseous and liquid particles of a substance suspended in air.

VERDIGRIS A gray-green layer that forms on copper when it is exposed to air.

VOLTAGE The force that pushes an electrical current around a circuit.

X-RAY A type of powerful electromagnetic radiation. X-rays can pass right through human tissue but not bone and are used in medical science to take internal pictures of the human body.